alkhder at AlexandriaX-2026: Context-Aware English-to-Dialectal Arabic Dialogue Translation with QLoRA-Fine-Tuned NileChat-3B

Hasan Alkhder
Sakarya University, Türkiye
hasan.alkhder2@ogr.sakarya.edu.tr

Maher Pasa
Altınbaş University, Türkiye
engmaherpasa@gmail.com

Maria Hanini
Imperial College London
m.hanini@imperial.ac.uk

Amro Najjar
Luxembourg Institute of Science and Technology
amro.najjar@list.lu

Ahmet Zengin
Sakarya University, Türkiye
azengin@sakarya.edu.tr

Abstract

This paper describes team Alkhder's submission to Subtask 1 of the AlexandriaX-2026 shared task, which asks systems to translate English dialogue turns into thirteen Arabic dialects. We fine-tune NileChat-3B-Base with QLoRA on the Alexandria parallel dialogue corpus, training one set of adapter weights for all dialects. Each turn is translated conditioned on the target country and dialect, the domain, the participant roles, the speaker–addressee gender direction, and the preceding dialogue context. At inference, conversations are translated autoregressively and the model is fed its own previous outputs back as context without using any gold target-side test information. Under the constrained track, using only Alexandria data, it trains in about one hour and translates the private test set in under two hours. The system reaches a country-macro-averaged spBLEU of 20.36 and chrF++ of 36.64 on both tracks.

1 Introduction

Dialectal Arabic machine translation (MT) remains difficult because Arabic varieties differ from Modern Standard Arabic (MSA) and from one another in vocabulary, morphology, syntax, pragmatics, and sociolinguistic use, including across cities within the same country (Abdul-Nabi et al., 2024). Progress is limited by scarce and domain-specific parallel data (Meftouh et al., 2015; Bouamor et al., 2018; Mohamed et al., 2023), non-standardized spelling (Alhafni et al., 2024; Kheder et al., 2024), and the need to use dialogue context and speaker information during translation (El Mekki et al., 2026; Bawden et al., 2016). The AlexandriaX-2026 shared task (El Mekki et al., 2026) addresses these issues by evaluating systems on meaning preservation and dialect-specific adaptation.

We participate in Subtask 1, *Context-Aware English-to-Dialectal Arabic Dialogue Translation*, in both the constrained and unconstrained track. The task is to translate selected English dialogue turns into one of thirteen dialectal Arabic varieties, using the available conversational and metadata context. Two varieties, Libyan and Sudanese, have no training data and are therefore evaluated in a zero-shot setting.

We build one system that satisfies the constrained track rules and submit it to both the constrained and unconstrained rankings. The system fine-tunes NileChat-3B-Base (El Mekki et al., 2025), a 3B-parameter Arabic and dialect-focused language model, with QLoRA (Dettmers et al., 2023) on turn-level prompts derived from the Alexandria corpus (EL Mekki et al., 2026). Each prompt includes the target dialect and the relevant dialogue metadata. One set of adapter weights is used for all dialects.[1]

We show that this compact approach trains in about one GPU-hour and covers all dialects. The system achieves a country-macro-averaged spBLEU of 20.36 and chrF++ of 36.64, ranking sixth in both tracks. We show that, as implemented, self-generated dialogue context contributes no measurable benefit. We further discuss the challenge imposed by the strong per-country resource imbalance in the training data against the country-macro-averaged evaluation metric.

The paper proceeds as follows. Section 2 covers the task setup and related work, Section 3 our system, and Section 4 the experimental configuration. Section 5 reports results, before concluding in Section 6.

2 Background

2.1 Task setup

AlexandriaX-2026 (El Mekki et al., 2026) includes three subtasks: context-aware English-to-dialectal Arabic dialogue translation, cross-dialect Arabic MT for short financial queries, and diagnostic

[1] Code: https://github.com/HasanAlkheder/AlexandriaX2026-Subtask1.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 864–871
October 28-29, 2026 ©2026 Association for Computational Linguistics

MT evaluation based on error spans and an LQM-inspired typology (Malaysha et al., 2024; Magdy et al., 2026). We participate in Subtask 1, which requires systems to translate selected English turns into a specified Arabic dialect using dialogue context, country and dialect labels, domain, persona descriptions, speaker–addressee gender direction and turn position. The translation must preserve meaning and use appropriate lexical, morphological, pragmatic and sociolinguistic forms. We enter both tracks. The constrained track allows only Alexandria data and models with up to 5B parameters, whereas the unconstrained track permits external data and models of any size.

2.2 Data

The task uses the Alexandria corpus (EL Mekki et al., 2026), which contains multi-domain, multi-turn English–dialectal Arabic conversations from thirteen countries. Training and development data are available for eleven countries and contain 66,480 and 12,250 turns, respectively. The data are highly imbalanced, with 14,933 Palestinian training turns compared with 2,034 Tunisian turns (Table 1 reports the train+dev pool). Libya and Sudan have no training or development data and are evaluated zero-shot. The public test set contains 14,442 turns, while the private test set contains 14,459 turns across 4,673 conversations, distributed near-uniformly over the thirteen countries (roughly 1,100 turns each, with 1,309 for Libya and 915 for Sudan) and scored through CodaBench.

Country	Available	Used (cap 2,500)
EG (Egypt)	4,221	2,500
JO (Jordan)	6,614	2,500
LB (Lebanon)	10,024	2,500
MA (Morocco)	3,683	2,500
MR (Mauritania)	6,629	2,500
OM (Oman)	7,389	2,500
PS (Palestine)	16,043	2,500
SA (Saudi Arabia)	9,580	2,500
SY (Syria)	7,190	2,500
TN (Tunisia)	3,150	2,500
YE (Yemen)	4,207	2,500
Total	78,730	27,500

Table 1: Turn-level training examples per country. Libya and Sudan have no training data.

2.3 Related Work

Early dialectal Arabic MT addressed limited data through crowdsourcing and MSA-based pivoting (Zbib et al., 2012; Salloum and Habash, 2013). Later resources such as PADIC, MADAR and AraBench expanded corpus coverage and evaluation, but remained limited by restricted domains and data scarcity (Meftouh et al., 2015; Bouamor et al., 2018; Sajjad et al., 2020). Arabic-specific pretrained models and instruction-tuned LLMs have improved performance, although results remain uneven across dialects (Kadaoui et al., 2023; El Mekki et al., 2025; Alkhder and Abboush, 2026; Jon et al., 2026).

Arabic dialects also lack standardized orthographies, which creates multiple valid spellings and affects automatic evaluation. Existing approaches normalize dialectal text into CODA or MSA (Habash et al., 2018; Alhafni et al., 2024; Khered et al., 2025). Context-aware MT further shows that previous dialogue turns improve coherence, pronoun resolution and ambiguity resolution, while speaker information such as gender helps reduce agreement errors (Tiedemann and Scherrer, 2017; Voita et al., 2018; Bawden et al., 2018; Vanmassenhove et al., 2018). AlexandriaX-2026 extends this work to English-to-dialectal Arabic translation across thirteen varieties by providing dialogue history and speaker–addressee gender metadata.

Recent low-resource MT systems adapt pretrained multilingual models (DeGenaro and Lupicki, 2024). Parameter-efficient methods such as LoRA and QLoRA make this feasible with limited computational resources (Hu et al., 2022; Dettmers et al., 2023).

3 System Overview

3.1 Prompt construction

Every training and test example is formatted as one instruction-style prompt with four blocks: (i) an **instruction** to translate only the final English turn into the named dialect, preserving meaning, tone, register, persona and gender agreement; (ii) a **metadata** block giving the target country and dialect name, domain, participant descriptions, and the current speaker with the gender direction (e.g. *female* $\rightarrow$ *male*); (iii) the **context**, i.e. the conversation so far truncated to the last four turns, where each previous turn appears as its English source and, when available, its Arabic translation; and (iv) the **query**, the English turn to translate, followed by a fixed response marker. Appendix A provides a prompt template.

The dialect is specified in natural language, allowing one adapter to serve all thirteen varieties,

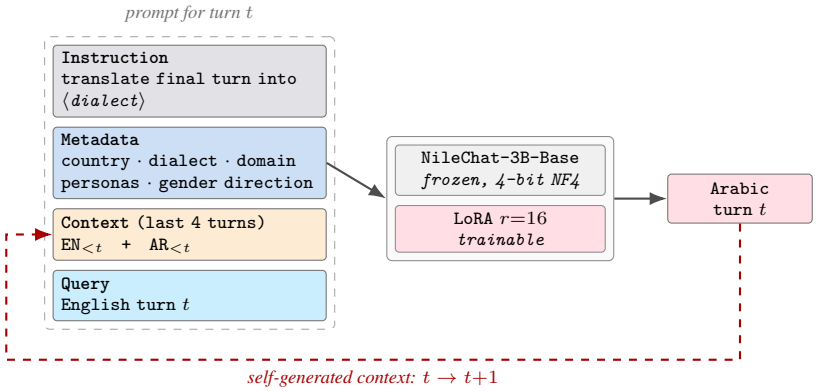


Figure 1: System overview.

including the zero-shot Libyan and Sudanese dialects. Although Libyan and Sudanese are absent from fine-tuning, the model can associate them with related varieties seen during pretraining, namely Maghrebi Arabic for Libyan and Egyptian Arabic for Sudanese. Training prompts use gold Arabic translations in the context, whereas inference prompts use the model's own previous outputs.

3.2 Base model and fine-tuning

As our base model, we select NileChat-3B-Base (El Mekki et al., 2025), a dialect-focused Arabic language model with ≈3B parameters leaving headroom under the 5B constrained track limit. We fine-tune with QLoRA (Dettmers et al., 2023). The base weights are frozen and quantised to 4-bit NF4 with double quantisation and bfloat16 compute. Low-rank adapters (Hu et al., 2022) with $r = 16$, $\alpha = 32$ and dropout 0.05 are trained on all attention and MLP projection matrices (`q/k/v/o_proj`, `gate/up/down_proj`). Training is run for one epoch using an adapter trained jointly on the eleven dialects for which training data are available. This also supports transfer across varieties and enables zero-shot evaluation for Libyan and Sudanese, for which no training data are available. Zero-shot varieties are reached at inference only, through the natural-language dialect label and the model's pretraining exposure to related varieties. Appendix B, Table 8 lists the configuration.

3.3 Context-aware inference

At test time, conversations are translated sequentially in dialogue order. Each Arabic translation the model produces for turn t is inserted into the context of turn $t + 1$. The dialogue context is therefore self-generated without using target-side information from the test set. This prevents leakage but could lead to error propagation since a mistranslated turn remains visible for up to four subsequent turns. Decoding is greedy (`num_beams`=1) with a repetition penalty of 1.05 and a cap of 200 new tokens. For throughput, at each turn position prompts from 16 conversations are batched with left padding and generated in parallel. All predictions are cached to persistent storage per country and the pipeline can be restarted after interruptions. Figure 1 summarizes our system.

4 Experimental Setup

4.1 Data preparation

We fold the dev split into training and create one supervised example for each dialogue turn. Each example contains the metadata, the preceding turns with their English sources and gold Arabic translations, and the gold Arabic translation of the current turn. This yields 78,730 examples (66,480 train + 12,250 dev). Because the official metric macro-averages over countries, we cap each country at 2,500 randomly sampled examples. The cap applies to all eleven varieties with training data and acts as a uniform downsample that equalises per-country counts (2,500 each, 27,500 in total) (Table 1).

4.2 Preprocessing and training

Prompts are limited to four preceding turns and truncated to 896 tokens, with older context removed first. We apply no orthographic normalization, dediacritization or transliteration because the Alexandria references preserve the original dialectal writing, including code-switched Latin-script tokens. We train for one epoch with a learning rate of 2e−4, cosine scheduling, 20 warmup steps, an effective batch size of 16, 1,719 optimizer steps, paged 8-bit AdamW and bf16 computation. Training uses one

NVIDIA A100-SXM4-80GB GPU and takes approximately one hour. Decoding the private test set takes approximately 1.5 hours.

4.3 Submission and evaluation

We submit predictions for every private-test turn in both tracks using the official JSON format. The files are validated to ensure complete coverage and no missing, duplicate or empty predictions before submission. The primary metric is country-macro-averaged spBLEU, with chrF++ reported as a secondary metric.

5 Results

5.1 Official results

Table 2 presents the official results. Our system achieves 20.36 spBLEU and 36.64 chrF++, ranking sixth in both the constrained and unconstrained tracks. The same predictions were evaluated in both tracks. Compared with the leading systems, we retain 80.7% of the chrF++ score but only 66.9% of the spBLEU score. Against the best unconstrained system, the figures are 76.1% and 60.8%. The chrF++/spBLEU ratio in Table 2 increases monotonically with rank in both tracks, suggesting outputs that remain similar at the character level but differ in inflection, clitic attachment, spelling or lexical choice.

#	Team	spBLEU	chrF++	Ratio
Constrained track				
1	Thakaa	30.42	45.42	1.49
2	LahjaMT	28.30	43.81	1.55
3	FCDS	26.48	42.30	1.60
4	Rosetta	26.10	41.79	1.60
5	Namaa Community	23.26	39.03	1.68
6	**Hasan_kh (ours)**	**20.36**	**36.64**	**1.80**
Unconstrained track				
1	Thakaa	33.49	48.13	1.44
2	cepin	32.50	47.05	1.45
3	LahjaMT	28.54	44.02	1.54
4	Namaa Community	27.41	42.58	1.55
5	Rosetta	25.09	41.02	1.63
6	**Hasan_kh (ours)**	**20.36**	**36.64**	**1.80**

Table 2: Official Subtask 1 results.

5.2 Post-submission analysis

These experiments were conducted post submission using matched public-test references and a fixed ablation set.

Code-switching. The system learns to code-switch in the varieties where the references use Latin-script material most frequently, especially Tunisian and Moroccan Arabic. However, its code-switching density is lower than the references in the main code-switching varieties, reaching 35–58% of the reference token rate (0.58 for Tunisian, 0.58 for Moroccan, 0.48 for Lebanese, 0.35 for Egyptian and 0.15 for Mauritanian; Table 3). We report percentage of turns containing at least one Latin-script token, percentage of all tokens in Latin script (token rates are Latin-script tokens divided by all whitespace-delimited tokens on that side's text), and the ratio of system to reference token rates.

	Turns (%)		Tokens (%)		
Country	**Ref**	**Sys**	**Ref**	**Sys**	**Tok ratio**
TN	47.5	32.3	6.88	3.97	0.58
MA	29.5	18.9	4.02	2.34	0.58
LB	21.2	10.4	2.32	1.12	0.48
EG	10.2	4.3	0.94	0.33	0.35
MR	7.0	1.3	0.74	0.11	0.15
SD	4.0	1.5	0.46	0.14	0.30
JO	4.1	3.4	0.86	0.32	0.38
SA	2.2	1.3	0.23	0.11	0.47
OM	2.1	1.2	0.23	0.10	0.42
PS	1.6	0.9	0.24	0.08	0.35
YE	1.0	0.5	0.10	0.05	0.53
LY	1.4	1.7	0.11	0.14	1.27
SY	0.8	1.3	0.08	0.13	1.63

Table 3: Latin-script code-switching on the matched public-test pair.

Replacing omitted loanwords with Arabic paraphrases reduces n-gram overlap and likely contributes to the lower spBLEU scores. In the two zero-shot varieties the direction reverses. The system over-produces Latin-script tokens relative to the references (1.27 for Libyan, 1.63 for Syrian). Under-production is therefore specific to the dialects whose references code-switch heavily, while the consistent relationship between the dialect label and code-switching density shows that the natural-language dialect label affects register-level behaviour, not only vocabulary.

Context ablation. We compared self-generated history, gold Arabic history and empty history on 200 conversations per country (Table 4).

Context condition	spBLEU	chrF++
B: gold Arabic history	21.62	37.66
C: empty history	21.41	37.23
A: self-generated (submitted)	20.56	36.73

Table 4: Context ablation on 200 conversations per country (8,000 turns), scored against public-test references.

Gold history achieves 21.62 macro spBLEU and 37.66 chrF++, while empty history achieves 21.41 and 37.23. The submitted self-generated condition performs worst, with 20.56 spBLEU and 36.73 chrF++. This shows that the model gains little from dialogue history and that errors in its previous translations can outweigh the benefits of context. The effect is consistent and empty history matches or outperforms the submitted self-generated condition in all thirteen varieties (Table 5).

Country	A: self	B: gold	C: empty	B−A	C−A
SY	26.88	27.48	28.08	+0.60	+1.20
JO	25.95	27.91	27.07	+1.96	+1.12
TN	25.29	26.38	26.32	+1.09	+1.03
SA	24.78	26.84	26.02	+2.06	+1.24
EG	23.80	24.81	24.43	+1.01	+0.63
PS	21.02	22.74	22.60	+1.72	+1.58
LB	20.99	21.46	21.68	+0.47	+0.69
OM	19.80	20.56	21.05	+0.76	+1.25
MA	19.30	19.85	20.00	+0.55	+0.70
YE	18.88	19.72	19.36	+0.84	+0.48
SD	16.13	17.27	16.91	+1.14	+0.78
LY	14.74	15.45	14.85	+0.71	+0.11
MR	9.75	10.55	9.94	+0.80	+0.19
Macro	**20.56**	**21.62**	**21.41**	**+1.06**	**+0.85**

Table 5: Per-country context ablation (macro spBLEU).

Orthographic normalization. Normalizing system outputs and references with five rulesrules—given in Appendix C—increases macro spBLEU from 20.80 to 22.36 and chrF++ from 37.01 to 38.28. The largest spBLEU improvements occur for Saudi (+2.68), Omani (+2.56), Sudanese (+2.11) and Egyptian (+2.08) Arabic, and the smallest for Moroccan (+0.64) and Syrian (+0.73) (Table 6, and per-country gains in Appendix D, Table 9). However, normalizing system outputs only, while leaving references unchanged, reduces macro spBLEU from 20.80 to 15.49 (-5.31). This suggests that the model already matches the spelling style of the corpus. Changing only the model outputs breaks matches with the references. Normalization is therefore only useful for analysis, when both the model outputs and the references are normalized in the same way. It should not be applied to the submitted outputs as it would lower the score.

	Raw	Normalised	Δ
spBLEU (macro)	20.80	22.36	+1.56
chrF++ (macro)	37.01	38.28	+1.27

Table 6: Effect of orthographic normalzation on the public-test evaluation.

MSA fallback. The system does not show a strong general tendency to fall back to MSA. Its MSA function-word rates are close to the reference rates, with only small positive residuals for Sudanese (+0.34), Tunisian (+0.35) and Mauritanian (+0.18) (Table 7). The apparent MSA signal in the raw system outputs therefore largely reflects the reference texts and not systematic model behavior.

Country	MSA Δ (system − reference)
TN	+0.35
SD	+0.34
MR	+0.18
EG	+0.16
SY	+0.09
LY	+0.09
JO	+0.08
OM	+0.07
SA	+0.04
LB	0.00
MA	−0.10
PS	−0.01
YE	+0.06

Table 7: MSA function-word rate difference.

Manual error analysis. We annotated 60 private-test turns, approximately five per country, using categories adapted from the shared task's LQM-inspired typology. Twenty-three turns were error-free (Appendix E, Table 10). Content errors were most common (19), followed by dialect–MSA confusion (11), which was concentrated in Libyan (3), Sudanese (3) and Mauritanian (2). Morphological agreement errors occurred in 8 turns across eight countries, while orthographic errors occurred in 2. Six further turns had understandable content but sounded non-native in register. The LLM annotator identified the content and morphology errors but missed these foreign-register cases, showing the value of human evaluation for dialectal naturalness.

6 Conclusion

We present a compact, dialect-focused 3B model fine-tuned with QLoRA on turn-level metadata-rich prompts. This provides a practical constrained track system for context-aware dialectal Arabic dialogue translation. The system trains in about one GPU-hour, feeds its own outputs back as context without any gold leakage at inference, and covers thirteen dialects. It achieves 20.36 spBLEU / 36.64 chrF++ on both tracks. Future work covers longer training with a larger LoRA rank and beam-search decoding.

References

Razan Abdul-Nabi, Rasha Obeidat, and Anas Bsoul. 2024. A survey on machine translation of low-resource arabic dialects. In *2024 15th International Conference on Information and Communication Systems (ICICS)*, pages 1–6. IEEE.

Bashar Alhafni, Sarah Al-Towaity, Ziyad Fawzy, Fatema Nassar, Fadhl Eryani, Houda Bouamor, and Nizar Habash. 2024. Exploiting dialect identification in automatic dialectal text normalization. In *Proceedings of The Second Arabic Natural Language Processing Conference*, pages 42–54.

Hasan Alkhder and Mohammad Abboush. 2026. Sdnlp at amiya 2026: Syrian arabic dialect modeling with lora. In *Proceedings of the 13th Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 359–364.

Rachel Bawden, Rico Sennrich, Alexandra Birch, and Barry Haddow. 2018. Evaluating discourse phenomena in neural machine translation. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1304–1313, New Orleans, Louisiana. Association for Computational Linguistics.

Rachel Bawden, Guillaume Wisniewski, and Hélène Maynard. 2016. Investigating gender adaptation for speech translation. In *Actes de la conférence conjointe JEP-TALN-RECITAL 2016. volume 2 : TALN (Posters)*, pages 490–497, Paris, France. AFCP - ATALA.

Houda Bouamor, Nizar Habash, Mohammad Salameh, Wajdi Zaghouani, Owen Rambow, Dana Abdulrahim, Ossama Obeid, Salam Khalifa, Fadhl Eryani, Alexander Erdmann, et al. 2018. The madar arabic dialect corpus and lexicon. In *Proceedings of the eleventh international conference on language resources and evaluation (LREC 2018)*.

Dan DeGenaro and Tom Lupicki. 2024. Experiments in mamba sequence modeling and NLLB-200 fine-tuning for low resource multilingual machine translation. In *Proceedings of the 4th Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP 2024)*, pages 188–194, Mexico City, Mexico. Association for Computational Linguistics.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *Advances in Neural Information Processing Systems*, 36.

Abdellah El Mekki, Houdaifa Atou, Omer Nacar, Shady Shehata, and Muhammad Abdul-Mageed. 2025. NileChat: Towards linguistically diverse and culturally aware LLMs for local communities. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10967–10991, Suzhou, China. Association for Computational Linguistics.

Abdellah El Mekki, AbdelRahim A. Elmadany, Samar M. Magdy, Saad Ezzini, Mo El-Haj, Mustafa Jarrar, Zaid Alyafeai, Bernard Ghanem, and Muhammad Abdul-Mageed. 2026. AlexandriaX 2026: The First Shared Task on Dialectal Arabic Machine Translation. In *Proceedings of The Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Abdellah EL Mekki, Samar M. Magdy, Houdaifa Atou, Ruwa AbuHweidi, Baraah Qawasmeh, Omer Nacar, Thikra Al-hibiri, Razan Saadie, Hamzah A. Alsayadi, Nadia Ghezaiel Hammouda, Alshima Mohammed Alkhazimi, Aya Hamod, Al-Yas Yaqoob Al-Ghafri, Wesam El-Sayed, Asila Ismail al Sharji, Mohamad Ballout, Anas Belfathi, Karim Ghaddar, Serry Sibaee, and 28 others. 2026. Alexandria: A multi-domain dialectal Arabic machine translation dataset for culturally inclusive and linguistically diverse LLMs. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 32567–32592, San Diego, California, United States. Association for Computational Linguistics.

Nizar Habash, Fadhl Eryani, Salam Khalifa, Owen Rambow, Dana Abdulrahim, Alexander Erdmann, Reem Faraj, Wajdi Zaghouani, Houda Bouamor, Nasser Zalmout, Sara Hassan, Faisal Al-Shargi, Sakhar Alkhereyf, Basma Abdulkareem, Ramy Eskander, Mohammad Salameh, and Hind Saddiki. 2018. Unified guidelines and resources for Arabic dialect orthography. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, Miyazaki, Japan. European Language Resources Association (ELRA).

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.

Josef Jon, Rawan Bondok, and Ondřej Bojar. 2026. Current state of LLMs for Arabic dialectal machine translation. In *Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script*, pages 329–363, Rabat, Morocco. Association for Computational Linguistics.

Karima Kadaoui, Samar M. Magdy, Abdul Waheed, Md Tawkat Islam Khondaker, Ahmed Oumar El-Shangiti, El Moatez Billah Nagoudi, and Muhammad Abdul-Mageed. 2023. TARJAMAT: Evaluation of bard and ChatGPT on machine translation of ten Arabic varieties. In *Proceedings of ArabicNLP 2023*, pages 52–75, Singapore (Hybrid). Association for Computational Linguistics.

Waad Ben Kheder, Josef Jon, André Beyer, Abdel Messaoudi, Rabea Affan, Claude Barras, Maxim Tychonov, and Jean-Luc Gauvain. 2024. Aladan at iwslt24 low-resource arabic dialectal speech translation task. In *Proceedings of the 21st International*

Conference on Spoken Language Translation (IWSLT 2024), pages 192–202.

Abdullah Khered, Youcef Benkhedda, and Riza Theresa Batista-Navarro. 2025. Dial2msa-verified: A multi-dialect arabic social media dataset for neural machine translation to modern standard arabic. In *Proceedings of the 4th Workshop on Arabic Corpus Linguistics (WACL-4)*, pages 50–62.

Samar M. Magdy, Fakhraddin Alwajih, Abdellah El Mekki, Wesam El Sayed, and Muhammad Abdul-Mageed. 2026. LQM: Linguistically motivated multidimensional quality metrics for machine translation. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 40470–40493, San Diego, California, United States. Association for Computational Linguistics.

Sanad Malaysha, Mo El-Haj, Saad Ezzini, Mohammed Khalilia, Mustafa Jarrar, Sultan Almujaiwel, Ismail Berrada, and Houda Bouamor. 2024. AraFinNLP 2024: The first Arabic financial NLP shared task. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 393–402, Bangkok, Thailand. Association for Computational Linguistics.

Karima Meftouh, Salima Harrat, Salma Jamoussi, Mourad Abbas, and Kamel Smaili. 2015. Machine translation experiments on padic: A parallel arabic dialect corpus. In *Proceedings of the 29th Pacific Asia conference on language, information and computation*, pages 26–34.

Najet Hadj Mohamed, Malak Rassem, Lifeng Han, and Goran Nenadic. 2023. Alphamwe-arabic: Arabic edition of multilingual parallel corpora with multiword expression annotations. In *Proceedings of the 14th International Conference on Recent Advances in Natural Language Processing*, pages 448–457.

Hassan Sajjad, Ahmed Abdelali, Nadir Durrani, and Fahim Dalvi. 2020. AraBench: Benchmarking dialectal Arabic-English machine translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5094–5107, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Wael Salloum and Nizar Habash. 2013. Dialectal Arabic to English machine translation: Pivoting through Modern Standard Arabic. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 348–358, Atlanta, Georgia. Association for Computational Linguistics.

Jörg Tiedemann and Yves Scherrer. 2017. Neural machine translation with extended context. *Preprint*, arXiv:1708.05943.

Eva Vanmassenhove, Christian Hardmeier, and Andy Way. 2018. Getting gender right in neural machine translation. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 3003–3008, Brussels, Belgium. Association for Computational Linguistics.

Elena Voita, Pavel Serdyukov, Rico Sennrich, and Ivan Titov. 2018. Context-aware neural machine translation learns anaphora resolution. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1264–1274, Melbourne, Australia. Association for Computational Linguistics.

Rabih Zbib, Erika Malchiodi, Jacob Devlin, David Stallard, Spyros Matsoukas, Richard Schwartz, John Makhoul, Omar Zaidan, and Chris Callison-Burch. 2012. Machine translation of arabic dialects. In *Proceedings of the 2012 conference of the north american chapter of the association for computational linguistics: Human language technologies*, pages 49–59.

A Prompt Template

Prompts are generated following the template below. Angle brackets mark fields filled from the dataset. In the actual prompt, `<arabic_translation_i>` contains the gold Arabic during training and the model's own previous output at inference.

```
Translate ONLY the last English turn
into <dialect_name>. Preserve the
meaning, tone, register, persona
and gender agreement of the speaker.
Output the translation and nothing
else.

### Metadata
Country: <country_name>
(<country_code>)
Dialect: <dialect_name>
Domain: <domain>
Participants: <persona_a>;
<persona_b>
Speaker: <speaker_role>
(<gender_direction>)

### Conversation so far
Turn <i> [<speaker_i>] EN:
<english_turn_i>
Turn <i> [<speaker_i>] AR:
<arabic_translation_i>
(last four turns only)

### Turn to translate
EN: <english_turn_t>
AR:
```

The model generates after the final marker. At inference, generation stops at the end-of-sequence token or after 200 new tokens.

B Full Configuration

Component	Setting
Base model	UBC-NLP/NileChat-3B-Base (≈3B params; limit 5B)
Quantisation	4-bit NF4, double quantisation, bf16 compute (bitsandbytes)
LoRA	$r = 16$, $\alpha = 32$, dropout 0.05; targets `q/k/v/o_proj`, `gate/up/down_proj`
Epochs	1
Learning rate	2e−4, cosine schedule, 20 warmup steps
Batch size	8 per device × 2 grad. accumulation (effective 16); 1,719 steps
Max seq. length	896 tokens (left truncation)
Context window	last 4 turns
Optimizer	paged AdamW 8-bit; bf16; gradient checkpointing disabled
Training data	27,500 examples (per-country cap 2,500; train + dev)
Decoding	greedy (`num_beams`=1), repetition penalty 1.05, max 200 new tokens
Inference batching	16 prompts, left padding
Hardware	1 × NVIDIA A100-SXM4-80GB (Colab); ≈1 h train, ≈1.5 h inference
Software	transformers 5.14.1, trl 1.9.0, peft 0.19.1, torch 2.11 (CUDA 12.8)

Table 8: System configuration

C Orthographic Normalization Rules

The following mappings are applied to both hypotheses and references before rescoring. Codepoints are given alongside the glyphs to avoid ambiguity.

- Alef variants أ (U+0623), إ (U+0625) and آ (U+0622) → bare alef ا (U+0627)
- Alef maqsura ى (U+0649) → ya ي (U+064A)
- Ta marbuta ة (U+0629) → ha ه (U+0647)
- Hamza carriers: waw with hamza ؤ (U+0624) → waw و (U+0648); ya with hamza ئ (U+0626) → ya ي (U+064A)
- Deletion of short-vowel and gemination diacritics ـَ ـُ ـِ ـّ ـْ ـً ـٌ ـٍ (U+064B–U+0652), superscript alef ـٰ (U+0670), and tatweel ـ (U+0640)
- Collapse of repeated whitespace

D Per-Country Results

	Public test		Norm.
Ctry	spBLEU	chrF++	Δ
SY	26.31	42.88	+0.73
JO	26.00	42.14	+1.33
TN	25.11	39.37	+0.79
SA	24.74	40.92	+2.68
EG	23.41	38.33	+2.08
LB	21.80	37.87	+1.65
PS	21.27	37.63	+1.58
OM	20.14	36.75	+2.56
SD*	18.69	34.89	+2.11
YE	18.48	35.16	+1.29
MA	18.02	34.71	+0.64
LY*	16.38	33.37	+1.19
MR	10.10	27.15	+1.57
Macro	20.80	37.01	+1.56

Table 9: Per-country results on the public test split. **Norm.** is the spBLEU gain from orthographic normalization. *zero-shot varieties.

E Annotation Protocol

We sampled 60 turns from our private-test predictions, stratified to give approximately five per country. Each turn was presented with its English source, the preceding dialogue context, the country and dialect label, and the speaker–addressee gender direction, but without a reference translation, since references for the private test set are not released.

Labels were assigned independently by one native-speaker annotator and by an LLM prompted with the same category definitions, and were then reconciled by the human annotator. Categories were adapted from the LQM-inspired typology of Subtask 3 (Magdy et al., 2026): content or meaning error, dialect–MSA confusion, morphological agreement error, orthographic error, and register or persona mismatch. A sixth category, *unnatural but content-acceptable*, was added by the human annotator during labelling for turns whose content was recoverable but whose register sounded non-native.

Category	Turns
Error-free	23
Content / meaning error	19
Dialect–MSA confusion	11
Morphological agreement	8
Unnatural, content acceptable	6
Orthographic error	2
Register / persona mismatch	1

Table 10: Annotation counts over 60 sampled turns.